

# Module 02: Agents — Philosophy of Mind and the Self-Model

**Course:** Active Inference 401 | **Unit:** Philosophical Foundations | **Audience:**  
Advanced undergraduates / graduate students

## Learning Objectives

1. Analyze the **self-model** in Active Inference and its relationship to philosophical theories of selfhood and personal identity.
2. Evaluate the FEP's position on the **hard problem of consciousness** and the explanatory gap.
3. Examine the concept of **agential minimalism** — what minimal conditions make something an agent?

## Key Concepts

### 1. The Self as Generative Model

Active Inference proposes that the "self" is a generative model that the organism maintains about its own states — the **self-model**:

- Interoceptive inference: the model of one's own bodily states (heart rate, hunger, emotion)
- Proprioceptive inference: the model of one's own body position and movement
- Narrative self: higher-order models about one's history, identity, and future plans

This connects to several philosophical traditions:

- **Hume's bundle theory:** The self is a bundle of perceptions — the FEP says it's a bundle of self-directed predictions
- **Metzinger's self-model theory:** The self is a virtual model generated by the brain — no homunculus needed

- **Buddhist no-self (anattā):** There is no fixed self, only impermanent processes — the FEP formalizes this as a continuously updated posterior

## 2. The Hard Problem and Active Inference

David Chalmers' **hard problem**: Why is there subjective experience (qualia) at all? Why doesn't inference happen "in the dark"?

Active Inference's response is nuanced:

- **What it explains:** The *structure* of experience — why perception has the character it does (governed by the generative model)
- **What it doesn't explain:** Why any subjective experience accompanies this processing
- **The modest claim:** Active Inference is a *process theory* that explains the *form* of consciousness, not its existence

Some theorists (e.g., Wiese & Metzinger, 2017) argue that Active Inference can address the hard problem through **phenomenal self-modeling** — consciousness is what it's like to be a system that models itself modeling the world. Others remain skeptical.

## 3. Agential Minimalism

What are the minimal conditions for agenthood?

The FEP suggests a **graduated ontology**:

- **Mere physical system:** Exists at nonequilibrium steady state; has a Markov Blanket
- **Simple agent:** Minimizes free energy through active inference (changes its environment)
- **Cognitive agent:** Maintains a hierarchical generative model with counterfactual depth
- **Self-conscious agent:** Models itself modeling the world (meta-cognition)

The philosophical question: Does a thermostat count as an "agent" under the FEP? If so, does this trivialize the concept of agency? This is the **dark room objection** applied to agency — if everything is an agent, nothing is.

## 4. Free Will and Determinism

Active Inference reframes the free will debate:

- **Determinism:** The agent's actions are determined by free energy minimization — there's no "uncaused cause"
- **Compatibilism:** Free will = acting according to one's own generative model (not external coercion). An agent is "free" when its actions flow from its own predictions and preferences
- **Autonomy:** Active Inference defines autonomy formally — a system is autonomous to the extent that its internal dynamics are driven by its own generative model rather than external perturbation

## 5. Personal Identity and Persistence

What makes you the "same" person over time? Active Inference offers a formal answer:

- **Structural continuity:** The generative model persists (with updates) across time
- **Narrative identity:** Higher-order models encode autobiographical memory and future plans
- **Allostatic regulation:** The organism maintains its essential variables within viable bounds — this is the *formal* criterion for persistence

## Summary

The self is a generative model the organism maintains about its own states. Active Inference offers a process account of selfhood that resonates with Humean bundle theory and Buddhist no-self. While it explains the structure of experience, it remains silent on the hard problem. Agential minimalism raises questions about the universality of agency, and the framework offers a compatibilist account of free will as action from one's own model.

## Further Reading

- Metzinger, T. (2003). *Being No One*. MIT Press.
- Seth, A. K. (2021). *Being You*. Dutton.

- Kirchhoff, M. D. & Kiverstein, J. (2019). *Extended Consciousness and Predictive Processing*. Routledge.
- Friston, K. J. (2018). Am I self-conscious? *Frontiers in Psychology*, 9, 579.